

NewsReader

*Building **structured event indexes** of large volumes of financial and economic data for decision making*

Luciano Serafini

FBK-IRST

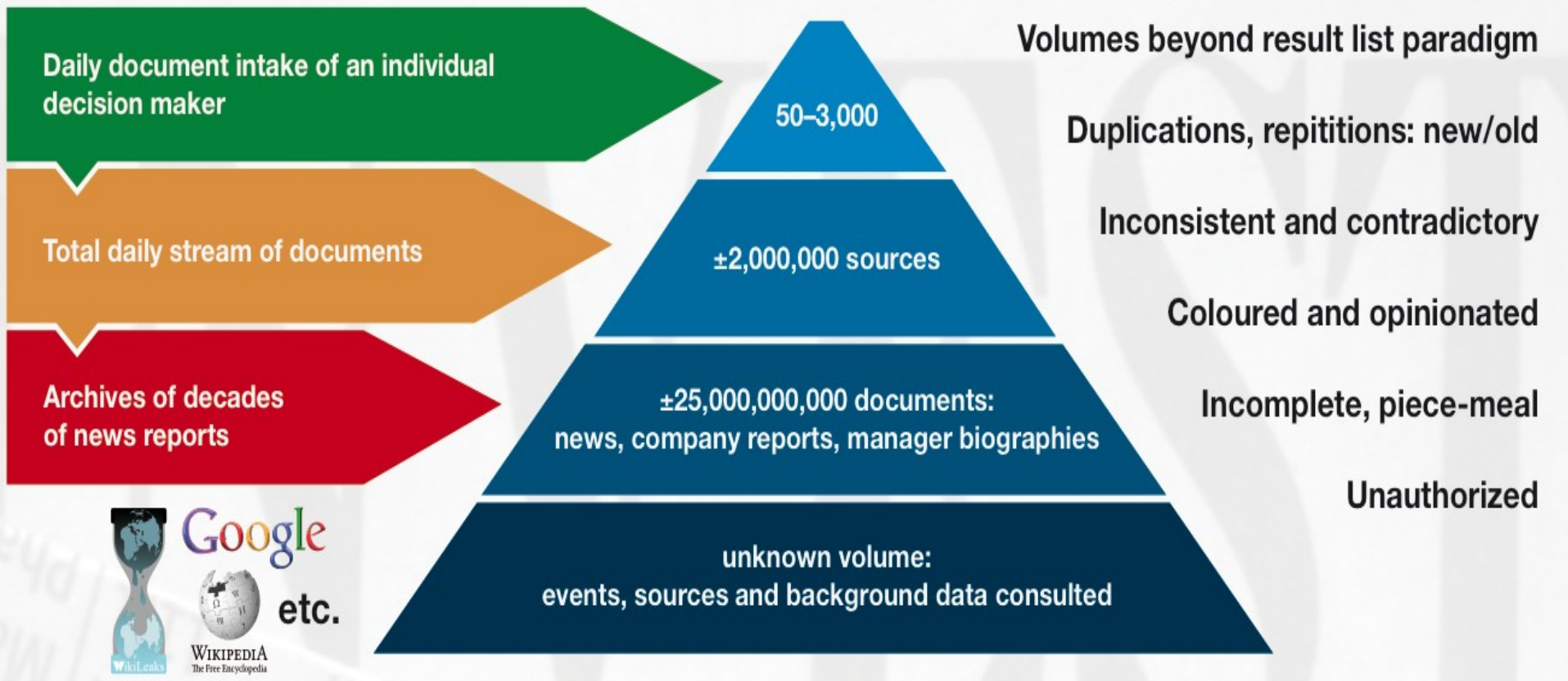
Trento, Italia

Motivations

- Professionals need to *have access to accurate and complete knowledge*
- *Increasing size of data* they need to consider
- *knowledge and information is evolving* and is quickly getting out of date
- They more and more *rely on technology that filter and select knowledge* and information because they cannot physically and mentally cope with it

Motivations

Decision maker information overload



Circa 806 risultati (0,28 secondi)



[Former **Porsche** chiefs charged with market manipulation over failed ...](#)

[Telegraph.co.uk](#) - 19/dic/2012

Prosecutors in Stuttgart said the former chief executive of **Porsche**, Wendelin Wiedeking, and former finance director, Holger Harter, had issued ...

[Former **Porsche** Executives Charged in Attempted VW Takeover](#)

[Fox Business](#) - 19/dic/2012

[Ex-**Porsche** CEO & CFO charged in failed **VW** takeover bid](#)

[WorldCarFans.com](#) - 19/dic/2012



Wall Street Jo...



The Guardian



Deutsche Welle



National Post



Victoria Time...



FRANCE 24



INAUTONEWS

[tutte le 122 fonti di notizie »](#)



Giornalettismo

[Porsche, l'ex ceo a processo per la scalata a **Volkswagen**](#)

[Il Sole 24 Ore](#) - 19/dic/2012

Porsche, l'ex ceo a processo per la scalata a **Volkswagen** ... automobilistica di Stoccarda tentò l'improbabile **takeover** del colosso di Wolfsburg.

[Porsche Wins New York Case Over **VW** Takeover After Appeals ...](#)

[Carscoop \(blog\)](#) - 28/dic/2012

The issue of **Porsche's** failed attempt to acquire **Volkswagen** in 2008 has surfaced again. Last week the automaker's former CEO Wendelin ...

[Porsche Fires "Italian Torpedo" Over Failed **VW** Takeover, Former ...](#)

[Carscoop \(blog\)](#) - 21/dic/2012

This is one of many cases in both Europe and the U.S. of shareholders suing **Porsche** for damages over its failed attempt to **take over VW** in

Tools with more semantics...

EMM News Explorer

PiratePad: c2k0o8ZsG6 EMM News Explorer

emm.newsexplorer.eu/NewsExplorer/home/it/latest.html

Most Visited FBK SSL VPN - H... People | FONDA... MyFBK

EMM NewsExplorer

News Analysis

RSS RSS feed for the latest news summary

Daily News Analysis, across languages and over time

Name Search Text Search

Menu principale

News Summary

About EMM NewsExplorer

News language and date

Language or country:

it - Italiano

Date:

Jan 2013

Mo	Tu	We	Th	Fr	Sa	Su
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Analysis over time

Clustered news for martedì 15 gennaio 2013

Read more...



Violenta figlia adottiva di 9 anni, 50enne arrestato a Cosenza
(AGI) - Cosenza, 15 gen. - I carabinieri hanno tratto in arresto un cinquantenne cosentino per violenza sessuale continuata aggravata nei confronti della figlia

View with Google Earth

Berlusconi: "Monti leaderino" E lancia Draghi al Colle
[37] de es fr nl pl sl hu

Nuovi attacchi del Cavaliere contro il professore 'protesi della sinistra': "Diverso da come pensavamo". Sulla scelta di tornare in campo: "Processo Mediaset...storie correlate Ruby, no dei giudici allo stop per le elezioni. La giovane marocchina non testimonia in aulaIngroia: "Nessun patto di desistenza col Pd".
repubblica 14.14.00 CET

Countries

- Venezuela (24)
- Francia (23)
- Turchia (22)
- Germania (19)
- India (17)
- Spagna (12)
- Egitto (11)
- Libia, Gran Giamahiria Libica Popolare Socialista (10)
- Cuba (8)
- Svezia (8)
- Belarus (7)
- Mali (6)
- Grecia (6)
- Svizzera (5)
- Emirati Arabi Uniti (5)
- Canada (5)
- Regno Unito (5)
- Kenya (5)
- Sud Africa (5)
- Afghanistan (5)
- Niger (4)
- Burkina Faso (4)
- Mauritania (4)
- Somalia (4)

This Week's New Stories

Berlusconi da Santoro: "Crisi? M... ho colpa" "Imu? Costretti a vota... Accuse ai professori
9 gennaio 2013 - 11 gennaio 2013
Liste clonate, la minaccia di Gril...
"Se non si risolve via dalle elezi...
11 gennaio 2013 - 15 gennaio 2013
Ruby, no dei giudici allo stop pe... elezioni. La giovane marocchina... testimonia in aula
13 gennaio 2013 - 14 gennaio 2013
Addio all'attrice Mariangela Mel... aveva 71 anni
11 gennaio 2013 - 12 gennaio 2013
Albertini invia un pizzino a...
Formigoni: «Se parlo io...». Salv... non gli fa onore. Il Pd: se sa, va... procura
11 gennaio 2013 - 14 gennaio 2013
Agli Oscar «Lincoln» fa il pieno... nomination. Sorprende «Amour... Michael Haneke
10 gennaio 2013 - 14 gennaio 2013
Bengasi, attentato al console... italiano. Scandalo contro la sua aut...
Read more...

This Month's New Stories

Venezuela, scompare aereo da

People

... actually, we are interested in
EVENTS and **STORIES**

- *Single events* and class of events:
 - How many **takeovers** happened during 2011 in Italy and among which companies?
- *Causal chains*
 - What are the **effects of the takeover** of Porsche by VolksWagen on the two companies?
- *Scripts and narratives* involving a set of entities
 - **How happened** that VW took over Porsche?

Volkswagen takeover 1,8M hits in Google

Textual Source

Following the *takeover* of Škoda Auto in 1991 by the Volkswagen Group

In late 2005, Porsche *took* an 18.65% *stake* in the Volkswagen Group, further *cementing their relationship*, and *preventing a takeover* of Volkswagen Group

On 26 March 2007, Porsche *took its holding* of Volkswagen AG shares to 30.9%, *triggering a takeover bid* under [German Law](#).

Porsche could *launch a full takeover bid* for Volkswagen, Europe's biggest car manufacturer, this week if the EU's highest court makes its widely expected decision to ban a post-war law *giving* the German state *effective control over* VW.

On 16 September 2008, Porsche *increased its shares* by another 4.89%, in effect *taking control of* the company, with more than 35% of the voting rights.

Hedge funds *face* Volkswagen *storm* as Porsche *takeover* boosts shares. VW shares have *risen sharply* this week as Porsche built a 75 per cent stake, and unveiled plans to force through a deal *to take control of* the Golf and Polo car maker.

Porsche AG *took step closer to controlling* the much larger Volkswagen AG by upping its share holdings to 50.8% in late Monday trading.

6 Jan 2009 – Porsche has been on *a quest to takeover* VW for more than two years.

With present economic conditions shrinking Porsche's available cash, the automaker may have *to adjust or delay its plans to gain full control of* Volkswagen. In January, Porsche *raised its stake* in Volkswagen to 50.76% gaining a majority stake.

29-June-2009 Porsche Rejects VW Takeover Offer. The *power struggle* between German automakers Porsche and Volkswagen escalated on Monday with Porsche rejecting VW's *takeover bid* as unfeasible.

23 Jul 2009 – Porsche Chief Executive Wendelin Wiedeking has *stepped aside* in a sign that Volkswagen *takeover of* its local rival is almost secured.

This is what we would like to have...

NewsReader

topic classification, parsing, named entity recognition,
matching with semantic lexicons, semantic extraction of events,
storyline detection, provenance storage, visualization for search and browsing

Story line visualization

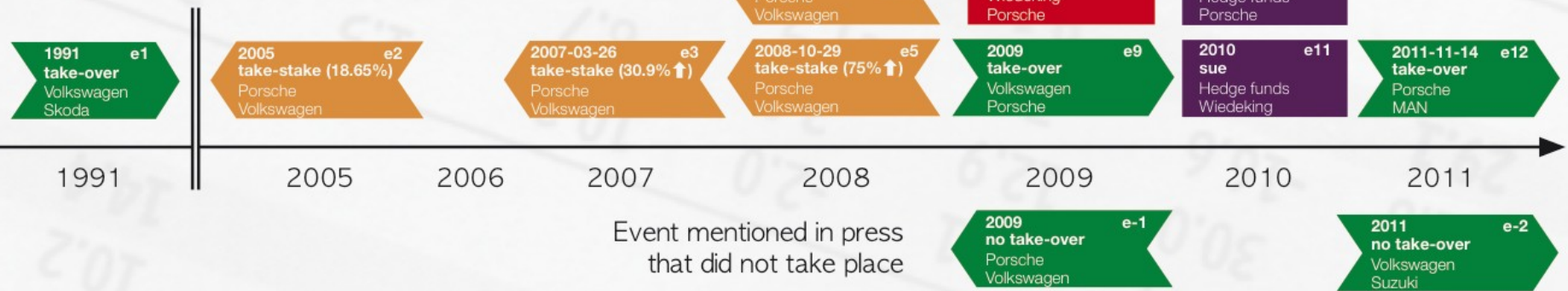
Entities

c1: Volkswagen
c2: Porsche
c3: Skoda
c4: Suzuki
c5: MAN
c6: Hedge funds
c7: Wendelin Wiedeking

Events

take-over: (e1, c1, c3), (e9, c1, c2), (e12, c1, c5),
(e-1, c2, c1), (e-2, c1, c4)
take-stake: (e2, c2, c1), (e3, c2, c1), (e4, c2, c1),
(e5, c2, c1), (e7, c1, c2)
sue: (e10, c6, c2), (e11, c6, c7)
step-aside: (e8, c7, c2)

Event mentioned in press
that actually took place



NewsReader will ...

- process streams of news in 4 different languages.
- extract **what happened to whom, when and where**,
 - **removing duplication** (event co-reference),
 - **complementing information** (event integration),
 - **registering inconsistencies** (reasoning about events)
 - **keeping track of the original** sources
 - **Distinguishing real events** from events that never happen (factual/hypothetical events)
 - **Discovering relations** between events (containment, temporal sequence, causation ...)
- archive everything in a **knowledge store**, which allows to retrieve and reason about all the acquired knowledge.

System Architecture

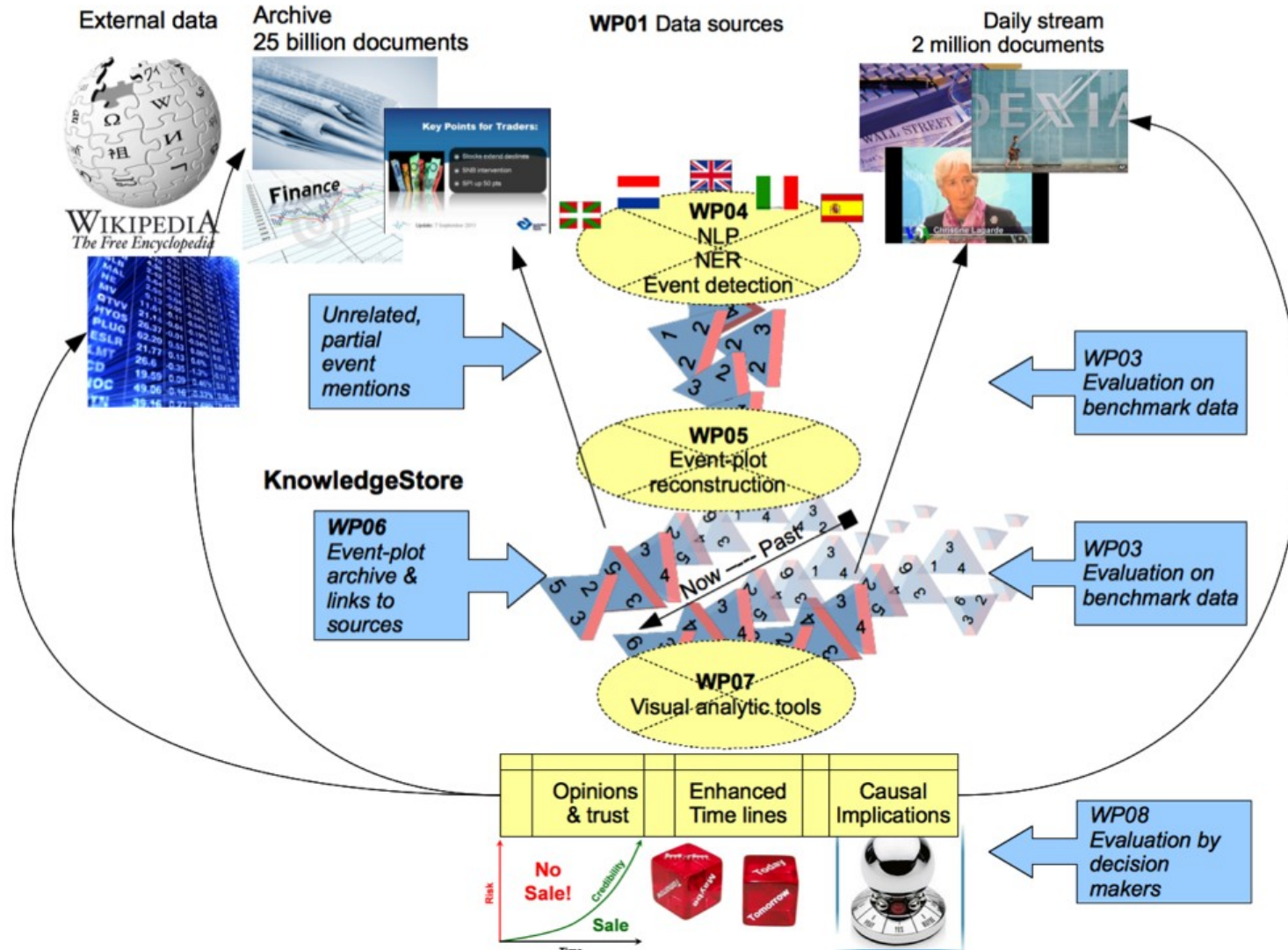


Figure 4: Overall project and system architecture

The Consortium



VU University
Amsterdam



FONDAZIONE
BRUNO KESSLER



eman ta zabal zazu

Universidad
del País Vasco

Euskal Herriko
Unibertsitatea



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